

Global and regional ocean carbon uptake and climate change: sensitivity to a substantial mitigation scenario

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Abstract Under future scenarios of business-as-usual emissions, the ocean storage of anthropogenic carbon is anticipated to decrease because of ocean chemistry constraints and positive feedbacks in the carbon-climate dynamics, whereas it is still unknown how the oceanic carbon cycle will respond to more substantial mitigation scenarios. To evaluate the natural system response to prescribed atmospheric “target” concentrations and assess the response of the ocean carbon pool to these values, 2 centennial projection simulations have been performed with an Earth System Model that includes a fully coupled carbon cycle, forced in one case with a mitigation scenario and the other with the SRES A1B scenario. End of century ocean uptake with the mitigation scenario is projected to return to the same magnitude of carbon fluxes as simulated in 1960

in the Pacific Ocean and to lower values in the Atlantic. With A1B, the major ocean basins are instead projected to decrease the capacity for carbon uptake globally as found with simpler carbon cycle models, while at the regional level the response is contrasting. The model indicates that the equatorial Pacific may increase the carbon uptake rates in both scenarios, owing to enhancement of the biological carbon pump evidenced by an increase in Net Community Production (NCP) following changes in the subsurface equatorial circulation and enhanced iron availability from extratropical regions. NCP is a proxy of the bulk organic carbon made available to the higher trophic levels and potentially exportable from the surface layers. The model results indicate that, besides the localized increase in the equatorial Pacific, the NCP of lower trophic levels in the northern Pacific and Atlantic oceans is projected to be halved with respect to the current climate under a substantial mitigation scenario at the end of the twenty-first century. It is thus suggested that changes due to cumulative carbon emissions up to present and the projected concentration pathways of aerosol in the next decades control the evolution of surface ocean biogeochemistry in the second half of this century more than the specific pathways of atmospheric CO₂ concentrations.

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1 Introduction

The ocean is the largest pool of inorganic carbon in dissolved form and on time scales of centuries to millennia controls the atmospheric levels of carbon dioxide (Raven

and Falkowski 1999; Sabine et al. 2004). The capability of the ocean to store part of the atmospheric carbon dioxide is called ocean uptake, which is driven by a positive surface-to-bottom gradient of total CO₂. This gradient is the result of the interplay between the physical processes of diffusive (turbulent) mixing and upwelling and the ocean carbon pumps. These pumps (Volk and Hoffert 1985) work to maintain a higher concentration of dissolved inorganic carbon (DIC) at depth than at the surface by means of biological (the soft-tissue and carbonate pumps) and chemical processes (the solubility pump). The growth of atmospheric CO₂ has been demonstrated to increase the ocean uptake, while at the same time the additional radiative forcing increases sea surface temperatures, thus reducing the efficiency of the solubility pump that is temperature dependent. The response of ocean biogeochemistry to an atmospheric CO₂ growth is less predictable, because a possible temperature-dependent increase in primary productivity may be contrasted by a concurrent increase in respiratory processes. Therefore, the net formation of organic carbon by biogeochemical processes may also be altered by possible changes in the supply of nutrients and in the food web structure. While there are indications that the biological pumps may not play a relevant role in the past and present uptake of anthropogenic carbon by the ocean (Gruber and Sarmiento 2002), the response under future emission scenarios may instead become relevant because of the changes in the ocean circulation underlying the functioning of biogeochemical processes (e.g. Sarmiento et al. 2004; Bopp et al. 2005).

The observed increase in fossil fuel utilizations, cement manufacturing and land-use changes in the current and previous century (Marland et al. 2008; Houghton 2008) has released substantial amounts of carbon dioxide in the atmosphere, which have not been redistributed among the ocean and terrestrial components as expected from geochemical considerations (Sarmiento and Gruber 2006, Chap. 10). In the span of the last 60 years, when direct instrumental data are available, the ocean and land have apparently been able to uptake only half of the carbon released from anthropogenic activities, as it can be derived by comparing the measured annual rate of change in atmospheric CO₂ concentration with the emission rate estimates (the so-called airborne fraction, Canadell et al. 2007; Marland et al. 2008; Houghton 2008). According to recent inverse model estimates, observational proxies and numerical simulations, the annual atmosphere-to-ocean global flux in the last 20 years is 2.2 ± 0.4 Pg C/y (Denman et al. 2007 and references therein), about one-third of the current estimates of fossil fuel emissions.

Results from state-of-the-art Earth System Models have shown that the major ocean basins feedback differently to a given business-as-usual climate change scenario (Crueger

et al. 2008; Frölicher and Joos 2010). Recent numerical simulations with forced ocean carbon cycle models indicate that the Southern Ocean may have already reduced its capability to uptake atmospheric CO₂ (Le Quéré et al. 2007; Lovenduski et al. 2008), although the scientific debate on what will be the uptake capability of the major oceans and the expected time scales is still open (e.g. Law et al. 2008; Zickfeld et al. 2008). In addition, the global and regional response of the future ocean carbon cycle to scenarios including stabilization conditions is still uncertain.

There is the need to assess and quantify the changes in the carbon uptake capacity of the different ocean basins under future scenarios including mitigation actions, specifically considering the role that ocean biogeochemistry may play by using more refined models as previously done (e.g. Sarmiento et al. 1998, 2004). This is of major interest because simple models cannot take into account the changes in the ocean carbon buffering capacity exerted by the presence of biogeochemical processes (e.g. Frankignoulle 1994).

The aim of this work is thus to investigate, by means of centennial simulations performed with a carbon cycle model, the large scale ocean processes involved in the exchange of carbon between the inorganic atmospheric reservoir and the inorganic and organic pools in the ocean. The focus is on the identification and evaluation of the exchange processes for a biogeochemical ocean system under strong external forcing: the increase in atmospheric CO₂ of the last part of the twentieth century and projected future increases, including mitigation scenarios. Therefore, atmospheric CO₂ concentrations instead of emissions are used to drive the carbon cycle model, following the simulation strategy proposed by Hibbard et al. (2007). This method partly reduces the uncertainties on the feedbacks of the natural ocean and land carbon pools and is well designed for stabilization experiments, because the achievement and maintenance of a target level of atmospheric CO₂ is likely to require large human intervention that is expected to be higher than the natural fluxes of carbon.

The future scenarios considered in this work are the A1B scenario and a substantial mitigation scenario. The A1B scenario is a SRES medium-high emission scenario without climate mitigation policy and driven by high economic growth, strong globalization, and rapid technology development (Nakicenovic and Swart 2000). The mitigation scenario is named E1, developed within the ENSEMBLES project with the aim of evaluating the effects of substantial mitigation actions (Den Elzen and van Vuuren 2007, Lowe et al. 2009) and very close to the RCP3-PD scenario proposed for CMIP5 (see Johns et al. 2011 for a scenario comparison). The motivation to run the A1B scenario is to compare the current results with those previously discussed in Meehl et al. (2007) and to contrast it with the E1 scenario.

This work focuses on the oceanic component of the carbon cycle with emphasis on the interaction between biogeochemistry and ocean circulation, and it is organized around the following scientific questions:

1. What is the role of the ocean in determining the allowed atmospheric emissions in order to match CO₂ target pathways in a model of the climate system that considers the feedbacks of fully dynamical natural carbon cycle?
2. What are the ocean physical and/or biogeochemical components that are more sensitive to changes in atmospheric CO₂ also in the case of a substantial mitigation effort?

In order to interpret the results of the ocean model within the larger perspective provided by the Earth System Model, prior to the presentation of the ocean model results (Sects. 4, 5), changes in the allowable global carbon emissions and in the global 2 m temperature are introduced in Sect. 3. The simulations have been performed with the INGV-CMCC-CE model (see Sect. 2) and are part of the ENSEMBLES multi-model experiment described in Johns et al. (2011), where information on the spread and uncertainty of the projections in meteorological parameters as well as in carbon fluxes are found. Discussion (Sect. 6) and summary and conclusion (Sect. 7) close the manuscript.

2 Methods

2.1 Model description

The INGV-CMCC Carbon cycle Earth system model (here after INGV-CMCC-CE) consists of an atmosphere–ocean–sea ice physical core coupled to models resolving carbon cycle on land and ocean, respectively. The technical description of the atmosphere–ocean coupling as well as the coupling of the carbon cycle models into the physical core are described in Fogli et al. (2009). The model components are: the ECHAM5 model for the atmosphere (Roeckner et al. 2006); the SILVA land surface model (Alessandri 2006; Alessandri et al. 2011); the OPA8.2 model for the ocean (Madec et al. 1998); the LIM2 model for the sea ice (Timmermann et al. 2005), and the PELAGOS model for the ocean biogeochemistry (Vichi et al. 2007a, b; Vichi and Masina 2009). The software used to couple the atmosphere (including the land-vegetation model) component and the ocean (including the biogeochemistry) is OASIS3 (Valcke 2006).

The ECHAM5 model numerically solves the primitive equations for the general circulation of the atmosphere on a sphere with state of the art physical parameterizations (see Roeckner et al. 2003, 2006). In this application, the

horizontal triangular truncation used is T31, while in the vertical 19 vertical levels and the top at 10 hPa are used. The OPA8.2 model is a primitive equation ocean general circulation model that is numerically solved on a global ocean curvilinear grid known as ORCA (Madec and Imbard 1996). In this application, we use ORCA2, with a resolution of 2 degrees of longitude and a variable mesh of 0.5–2 degrees of latitudes from the equator to the poles. The vertical grid has 31 levels (the 31st level is below the bottom) with variable layer depth and a constant 10 m step in the top 100 m. The atmosphere is coupled to the ocean and sea ice models with a coupling step of one day and the exchanged fields and coupling procedures are fully detailed in Fogli et al. (2009). With respect to the previous ECHAM/OPA coupled model (INGV-SXG, Gualdi et al. 2008), in the coupling interface with OPA8.2, here it is made use of the capability of ECHAM5 to compute surface heat fluxes for both the ocean and sea ice surfaces at the same grid point and then to combine them according to an ocean and sea ice fractional mask (see Fogli et al. 2009, for the detailed description of the technical implementation). The coupling time step between the LIM2 sea ice model and the OPA8.2 ocean model is 8 h. The ocean state variables are accumulated and averaged at every ocean-sea ice coupling time step.

The Surface Interactive Land Vegetation model (SILVA, Alessandri 2006; Alessandri et al. 2011) simulates land surface processes and their associated variability. The biophysical version of the SILVA model, which includes also carbon and vegetation dynamics, is discussed in Alessandri et al. (2011). The vegetation and carbon dynamics and the CO₂ flux exchange are derived from the core parameterizations of VEGAS (VEgetation-Global-Atmosphere-Soil, Zeng et al. 2004).

The PELAGOS model (PELAGic biogeochemistry for Global Ocean Simulations model, Vichi et al. 2007a, b) has been further extended to incorporate a full description of the dissolved inorganic carbon (DIC) dynamics and adequately simulate the ocean components of the carbon cycle. The PELAGOS model consists of the global ocean version of the Biogeochemical Flux Model (BFM, <http://www.bfm.cmcc.it>) and its coupling to the OPA OGCM. A detailed assessment of an interannual forced simulation of the last 20 years of the twentieth century is presented in Vichi and Masina (2009) with a particular focus on the evaluation against existing datasets of carbon production and transfer through the lower trophic levels of the marine food web. For the current application, the model also includes dynamics of dissolved inorganic carbon and the computation of surface exchange fluxes according to the carbonate chemistry equations (e.g. Zeebe and Wolf-Gladrow 2001) and air–sea gas transfer interactions (Wanninkhof 1992).

The PELAGOS model is written in a generalized mathematical formulation that allows the description of lower trophic levels and major inorganic and organic components of the marine ecosystem from a unified functional perspective (Vichi et al. 2007a). The pelagic state variables of PELAGOS are three unicellular planktonic autotrophs (picophytoplankton, nanophytoplankton and diatoms), three zooplankton groups (nano-, micro- and meso-) and bacterioplankton. The other chemical functional families are nitrate, ammonium, orthophosphate, silicate, dissolved bioavailable iron, oxygen, carbon dioxide and dissolved and particulate (non-living) organic matter (DOM, POM), for a total of 44 state variables. The implementation of carbonate chemistry for the closure of the carbon cycle adds 2 dynamically transported variables (total alkalinity and total dissolved inorganic carbon) and 5 diagnostic variables for the carbonate speciation (CO_2 , bicarbonate and carbonate concentrations, CO_2 partial pressure and pH). When coupled to the ocean model, PELAGOS is computed every 4 time steps of the ocean model.

2.2 Experiment set up

The experimental set up follows the multi-model ENSEMBLES concerted experiment described in Johns et al. (2011). Namely, with the INGV-CMCC-CE model the following simulations have been performed and are here reported: an historical (1860–1999) run and two future scenario runs. The historical simulation started from a 300 years physics-only control run at pre-industrial conditions, specifically with well mixed greenhouse gases (GHG: CO_2 , CH_4 and N_2O), ozone and sulphate aerosols fixed at 1860. The initialization of the pre-industrial run followed Stouffer et al. (2004) and was started from historical oceanic conditions representative of current temperature and salinity distributions (Levitus et al. 1998). During the physical control run the coupled model adjusted to the radiative fluxes of a preindustrial atmosphere. The control run is characterized by a weak negative trend in surface temperature (about -0.2°C per century), which was not removed from the results presented in this work.

Land and ocean carbon pools were initialized independently using dedicated acceleration techniques. Oceanic DIC and alkalinity pools have been initialized from current climate data reconstructions (Key et al. 2004) and DIC has been spun up to equilibrium with the preindustrial atmospheric CO_2 concentration by means of an acceleration method (adapted from Alessandri 2006 and Alessandri et al. 2011) consisting of increasing the air–sea CO_2 outgassing flux of a factor 20 and removing the corresponding DIC amount homogeneously from the oceanic pool. This procedure accelerates the release of the anthropogenic

oceanic carbon to the atmosphere and ensures the preservation of the vertical gradients observed in the current climate. It has been found that the application of the acceleration procedure for a period of 100 years was sufficient to reach dynamical equilibrium. Thereafter, once the dynamical equilibrium is reached, the initial oceanic DIC pool is found to be reduced of about 300 Pg C, a value that is slightly more than half the cumulative carbon emissions estimated from fossil fuels and land-use change (Marland et al. 2008; Houghton 2008). The terrestrial carbon pools have been initialized from scratch (empty pools), following the acceleration method of Alessandri (2006) over a few hundreds years. The preindustrial run with the full carbon cycle has been prolonged for another 100 years to provide a control reference. Nutrients are initialized from contemporary climatological datasets (Conkright et al. 2002) and the simulations were performed without any restoration to the background values. A constant decreasing linear trend in total nitrogen and phosphorus of 0.3% every 10 years has been removed from the results presented in this work.

The well-mixed GHGs and the sulphate aerosols used in the preindustrial and historical runs are the annually prescribed observation-based concentrations available from the ENSEMBLES multi-model experiment (Johns et al. 2011; forcing data are publicly available at http://www.cnrm.meteo.fr/ensembles/public/model_simulation.html). The future scenario experiments respectively employed the A1B SRES (Nakicenovic and Swart 2000) and E1 anthropogenic scenarios for the well-mixed GHGs and the sulphate aerosols, the latter developed with the IMAGE 2.4 Integrated Assessment Model (IAM, MNP 2006; van Vuuren et al. 2007; Lowe et al. 2009). The INGV-CMCC-CE simulations were performed without any variation in natural forcings (solar and volcanic).

The time evolution of the atmospheric CO_2 concentrations and total sulphur burden used are shown in Fig. 1a. By 2025, the A1B and E1 scenarios have already diverged. The CO_2 concentration in E1 peaks around 2040 and then slowly decreases for the whole second part of the twenty-first century. The concentration of sulphate aerosol for A1B is the same used in previous simulations (Boucher and Pham 2002) and reported in Meehl et al. (2007), while the new field concentrations for E1 were specifically computed (with the same chemistry climate model) for the ENSEMBLES multi-model experiment (Johns et al. 2011). The A1B SRES scenario shows a strong increase in the first quarter of the twenty-first century with a peak in 2020 and a rapid decline afterward. E1 is characterized by an immediate decline reaching a plateau after 2050. The ozone distribution from 1860 to 2100 is based on Kiehl et al. (1999), and includes the tropospheric ozone increase in the recent past decades, stratospheric ozone depletion and a simple projection of stratospheric ozone recovery.

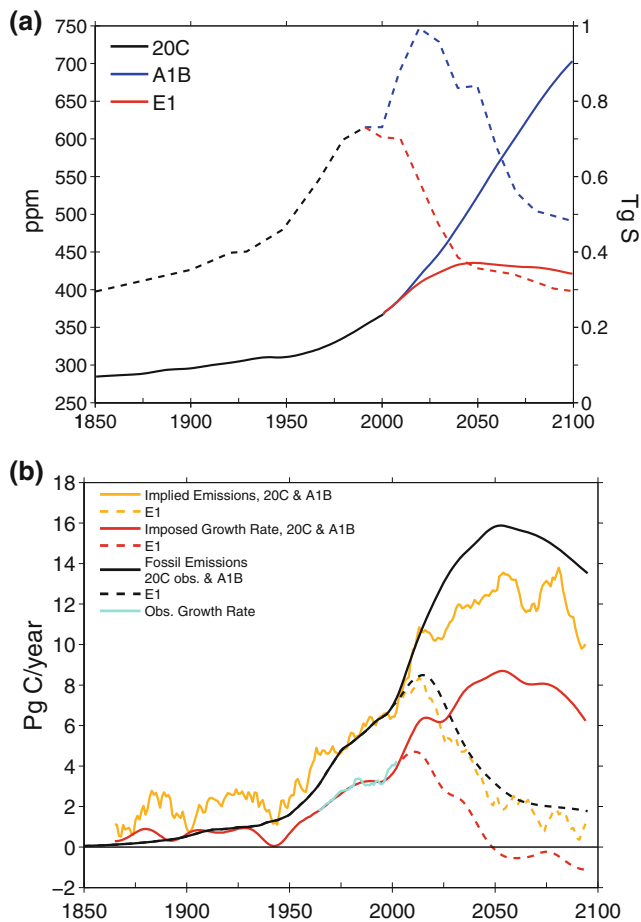


Fig. 1 Time evolutions of **a** prescribed atmospheric CO₂ concentrations (left axis, continuous lines) and total sulphate burden (right axis, dashed lines) and **b** observed and simulated carbon fluxes in the 20C and projections for the A1B and E1 scenarios. Twenty-first century scenarios of fossil emissions have been obtained with the IMAGE integrated assessment model (van Vuuren et al. 2007; Lowe et al. 2009). Data are shown with a 11-year running mean

Although the atmospheric CO₂ concentration was prescribed from observations for the whole historical simulation (1860–1999), previously of ~1950 the atmospheric CO₂ concentration is not dominated by anthropogenic emissions, while natural variability might have played a substantial role (see also Johns et al. 2011). It is argued that the Hibbard et al. (2007) design is questionable for the pre-1950 period, because the imposed CO₂ growth rate derived from atmospheric CO₂ proxy estimates of the early part of the historical period is likely to be affected by natural carbon cycle oscillations associated with internal climate variability and therefore inconsistent with the modelled internal climate variability. Most of the presented results therefore focus on the 1960–2100 period, to avoid these ambiguities in the interpretation of the results.

By specifying the time evolution of the atmospheric CO₂ concentration in a model with land and ocean prognostic carbon (e.g. a carbon cycle model), it is possible, on

average, to calculate the anthropogenic global emissions that would be necessary to balance the modelled land-to-atmosphere and ocean-to-atmosphere natural carbon, by subtracting to the specified atmospheric CO₂ growth rate the diagnosed natural carbon fluxes from the model. The anthropogenic emissions so estimated (orange curve in Fig. 1b) are defined “implied carbon emissions” and can be compared with the results of the Integrated Assessment Models (IAM) used to produce the target concentration pathways of GHGs. In Johns et al. (2011) the implied emissions from a number of carbon cycle models are compared. Here the analysis is focused on the contribution of the ocean carbon cycle model to the determination of the implied carbon emissions.

3 Global changes: historical simulation and projections

Figure 1b shows the time evolution of carbon fluxes from 1850 to 2100. Both the A1B and E1 projections are shown. For the 1960–2000 period, the implied emission simulated by the INGV-CMCC model falls close to the anthropogenic emission estimates (Marland et al. 2008). The computation of implied fluxes in the twentieth century represents the validation of the model capability to capture the land–ocean–atmosphere feedbacks in the carbon cycle over this period.

For the future scenarios, model results are comparable to those made with the IMAGE IAM (van Vuuren et al. 2007; Lowe et al. 2009). The imposed atmospheric CO₂ growth rate for the 1960–2000 period is derived from the observations (Johns et al. 2011), hence the excellent agreement with the observed one, while for the scenarios is derived from the specified atmospheric CO₂ evolutions, A1B and E1 respectively. Figure 1b shows that the A1B and E1 implied carbon emissions diverge as soon as the atmospheric CO₂ growth rate does so, around 2010, indicative of a relatively fast response of the natural carbon fluxes to the changes in atmospheric CO₂ concentration. Estimating an uncertainty in the model results of 1.5 Pg C/year (from Johns et al. 2011, but also here from the difference between INGV-CMCC-CE and IMAGE), Fig. 1 shows that in order to achieve the E1 atmospheric CO₂ stabilization, by 2040–2060 the carbon emission will have to be reduced to the level of about 1960, i.e. those of a century before. The comparison of the INGV-CMCC-CE model with the rest of the ENSEMBLES models discussed in Johns et al. (2011) shows that these results are within the models’ range, and typically close to the multi-model mean.

In both scenario simulations, the implied flux is lower than the estimate from the IMAGE model (Fig. 1b). The permissible emissions computed by the INGV-CMCC-CE model for A1B are required to be about 18% lower than the simpler IMAGE model. In the E1 scenario the permissible

emissions are also lower than estimated with the IMAGE model, although this number is the net result of oscillations mostly caused by the terrestrial fluxes that in this scenario are of the same order of the ocean uptake (not shown).

Figure 2 shows the time evolution of the annual global average of the surface temperature anomaly, from 1960 to 2100. For the 1960–2000 period, the temperature anomalies simulated by the INGV-CMCC-CE model compare well with those estimated by reanalysis data (ERA40, Uppala et al. 2005), in accord with previous works (Meehl et al. 2007). During the first half of the twenty-first century both the A1B and E1 projected temperature steadily increases, because in the early part of the century the warming is due to the ocean inertia responding to the atmospheric CO₂ already present in the atmosphere at 2000 (Meehl et al. 2007). In addition, the temperature projection is higher in E1 than A1B during the first part of the twenty-first century because the sulphate aerosol burden is decreasing throughout the twenty-first century in E1, while it is peaking in 2020 in A1B (Fig. 1a). This additional radiative forcing in a cooling sense has been diagnosed in all the models used for these ENSEMBLES simulations (Johns et al. 2011). Around mid twenty-first century, the E1 and A1B temperature projections diverge; clearly, the rate of increase of the temperature anomaly is substantially decreased in the E1 scenario in the second part of the twenty-first century and the E1 temperature anomaly remains below 2°C for all the century (with respect to the 1980–1999 average).

The surface carbon fluxes with the ocean component are presented in Fig. 3 for the control (preindustrial), historical and scenario simulations. The ocean is the largest carbon sink for anthropogenic emissions in the model, as shown in

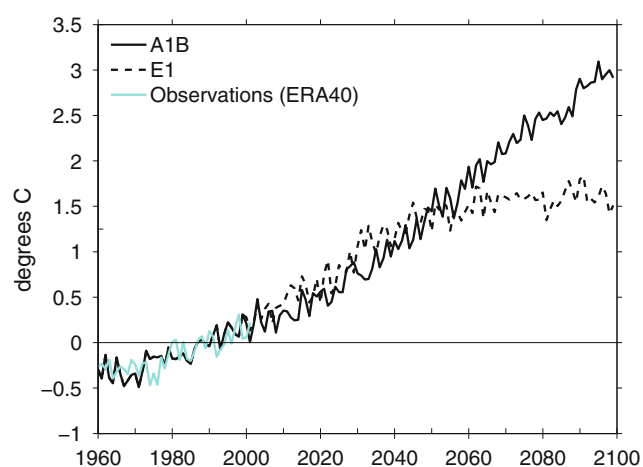


Fig. 2 Globally-averaged observed and simulated surface temperature anomalies (annual mean anomalies from the 1980–1999 climatology) in the 20C and projections for the A1B and E1 scenarios. Observations are from the ERA40 dataset and cover the period 1960–2001

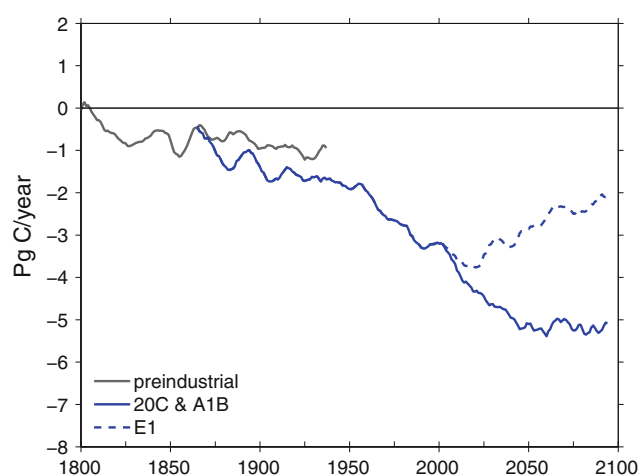


Fig. 3 Globally-averaged ocean carbon fluxes with the atmosphere in the control (preindustrial), 20C and scenario simulations (A1B and E1, 11-year running means). The superimposed control scenario is the continuation of the model spinup with constant preindustrial atmospheric aerosol, GHGs, and ozone concentrations. Fluxes are negative from the atmosphere to the ocean components

Figure 12 of Johns et al. (2011) and also deducible from the allowable global carbon emissions and the imposed atmospheric CO₂ growth rate shown in Fig. 1b. Note also that the large oscillations found during the twentieth century and at the end of the twenty-first in the allowable global carbon emissions (Fig. 1b) are mostly due to land exchanges, given the lack of such oscillations in Fig. 3. During the preindustrial simulation the ocean CO₂ fluxes adjust to a negative background value of about -1 Pg C/year with. The ocean uptake becomes constant after 2060 in the A1B scenario whereas in the E1 scenario it slowly decreases to the values of 1960 after peaking in 2020. The simulated current oceanic carbon uptake is larger than reported by other authors with different methodologies for the last period of the twentieth century (e.g. Gruber et al. 2009; Denman et al. 2007).

4 Regional validation: historical simulation

4.1 SST

Mean annual fields of Sea Surface Temperature (SST) in the 20C simulation (1970–1999) and differences with the A1B and E1 scenarios are presented in Fig. 4. The global mean SST is slightly colder on average than that calculated from HadISST data (Rayner et al. 2003), specifically the modelled global mean bias is -1.42°C . This cold bias is mostly confined to the middle–high latitudes (bias = -1.62°C), being reduced to -0.76°C in the tropics (30°S – 30°N) and it does not affect the structure of the SST. The overall assessment of the annual climatology in modelled SST versus HadISST data leads to an unbiased root mean

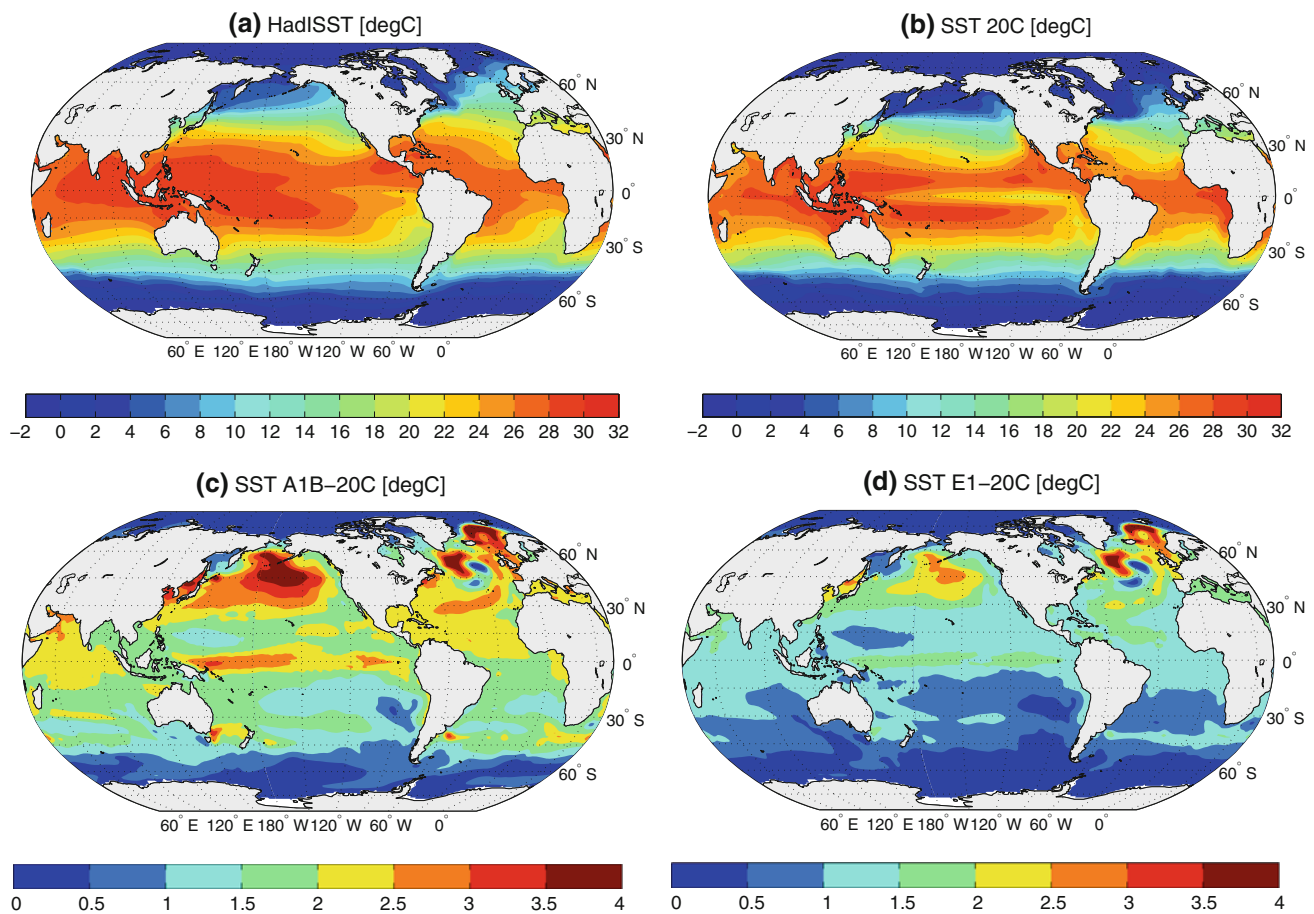


Fig. 4 Maps of the mean annual SST distribution during the twentieth century from **a** HadISST dataset, **b** 20C simulation (1970–1999) and differences with the scenarios: **b** A1B (2070–2099) – 20C (1970–1999); **c** E1 (2070–2099) – 20C

square error of 1.78°C and linear correlation of 0.98, which indicates a good description of the major SST patterns. The diagnosed bias in modelled SST is comparable to typical biases in state-of-the art climate models (Randall et al. 2007).

4.2 Meridional overturning circulation

The mean streamfunction of the meridional overturning circulation (MOC) in the Atlantic Ocean is presented in Fig. 5a. The streamfunction pattern agrees well with other coupled climate models (e.g. Crueger et al. 2008) and model-based estimates of historical MOC (Lozier et al. 2010). The maximum intensity of the simulated MOC is however on the lower end of the range of IPCC models presented in Schmittner et al. (2005) and Meehl et al. (2007), which is attributable to the low resolution of the atmospheric model (Marti et al. 2010 reported an increase from 11 to 15 Sv using the same ocean model used in this work but doubling the atmospheric resolution).

4.3 Anthropogenic carbon inventory

It has been estimated that a substantial portion of the anthropogenic carbon emitted to the atmosphere has been taken up by the ocean regions with higher ventilation rates (Sabine et al. 2004). The distribution of the anthropogenic carbon inventory in the global ocean, defined as the difference in DIC concentration between current climate conditions and the preindustrial ocean, is therefore one possible indicator of the model ability to simulate the ocean carbon cycle dynamics. The deep storage of anthropogenic carbon is closely related to the intensity of the thermohaline circulation.

The estimated distribution of anthropogenic carbon closely follows the pattern of the thermohaline circulation in the North Atlantic (Fig. 5b). This figure agrees well with the estimates of Sabine et al. (2004) and Feely et al. (2004) derived from in situ data, which shows that the highest concentrations of anthropogenic CO_2 are found in the surface waters. The mixing rate in the southern Atlantic gyre is apparently higher than estimated by Sabine et al. (2004)

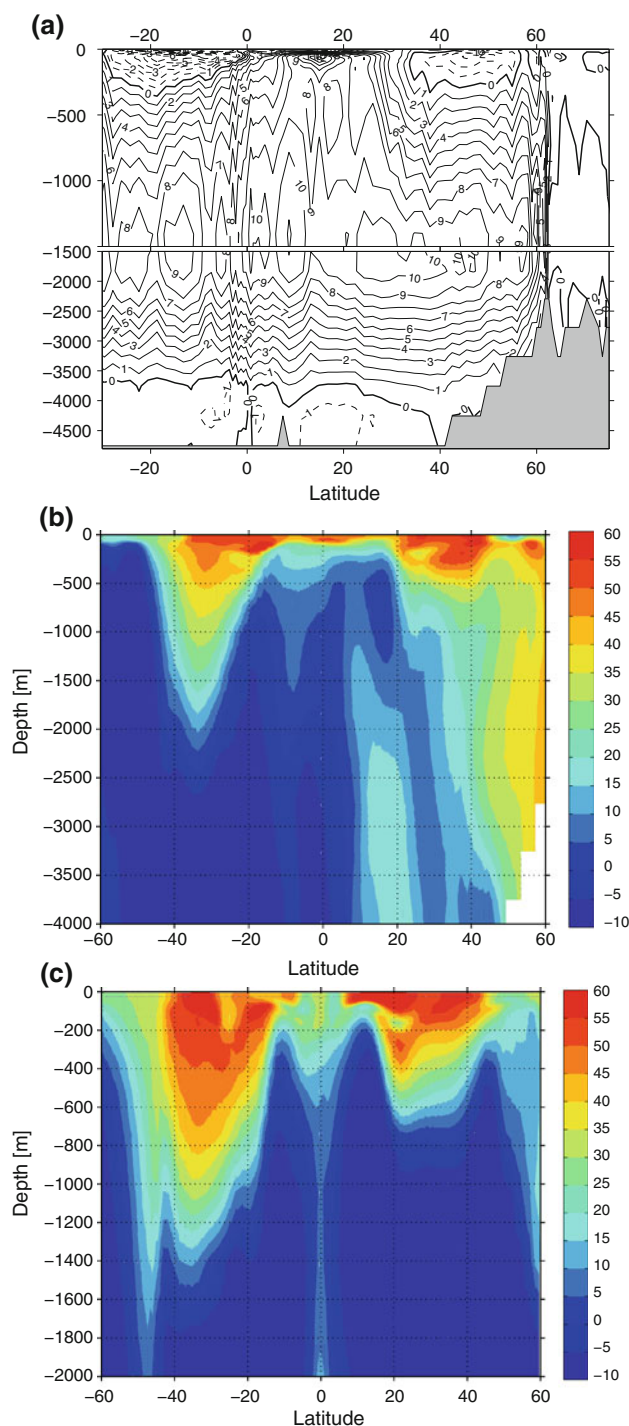


Fig. 5 a Mean streamfunction of the meridional overturning circulation in the Atlantic Ocean for the 20C simulation (1970–1999, in $S_{\text{verdrup}} = 10^6 \text{ m}^3 \text{ s}^{-1}$) and sections of the estimated anthropogenic carbon inventory in **b** Atlantic and **c** Pacific for the same time period (in $\mu\text{mol C kg}^{-1}$)

since the anthropogenic carbon has penetrated deeper in the water column. This may also occur due to a too intense convergence of the DIC-rich subpolar waters.

A similar feature is also found in the southern Pacific Ocean and partly in the northern Pacific (Fig. 5c). The equatorial upwelling is marked by a low concentration of anthropogenic CO_2 as found in data estimates, although the model also reveals that part of the anthropogenic carbon may have been exported at depth thanks to the sinking of organic particles and subsequent remineralization. The lack of this pattern in Sabine et al. (2004) may be attributed to the low spatial and temporal resolution of the used dataset, although a similar maximum of anthropogenic carbon at depth is visible in the Pacific section presented by Gruber et al. (2009, their Fig. 5).

4.4 CO_2 partial pressure

The surface partial pressure of CO_2 ($p\text{CO}_2$) determines the exchange of inorganic carbon with the atmosphere according to empirical parameterizations (e.g. Wanninkhof 1992). It is thus one of the carbon cycle variables that can be used to partly assess the simulated carbon fluxes under current climate conditions. Surface $p\text{CO}_2$ data shown in Fig. 6 have been obtained from the LDEO dataset (Takahashi et al. 2009, covering the period 1968–2008) and binned onto a regular 2×2 degrees grid to obtain an annual distribution to be compared with model annual means of the twentieth century. Model data have been interpolated onto the same grid excluding bins with missing data in the LDEO dataset. Figure 6 shows that the model reproduces the pattern of high $p\text{CO}_2$ in the large scale equatorial upwelling of the Pacific Ocean and low values in the Southern Ocean. The minima in the northern Pacific subtropical gyre are lower than observed as well as in the Kuroshio region, due to a previously known bias of higher primary production and particle export as it was found in forced simulations as well (Vichi and Masina 2009).

Major model biases are found in the equatorial Pacific where the upwelling waters are characterised by lower values and in the tropical Indian and Atlantic Oceans where the regions of high $p\text{CO}_2$ are absent, indicating that these latter areas act as sinks and not as sources in the historical simulation (see also Sect. 5.2). This bias is attributed to the coarse resolution of the ocean model (also found in forced simulations, e.g. Vichi et al. 2007b; Vichi and Masina 2009). The upwelling simulated by the physical model involves only surface waters which are progressively depleted of dissolved inorganic carbon by primary production instead of being resupplied by deep DIC-rich waters.

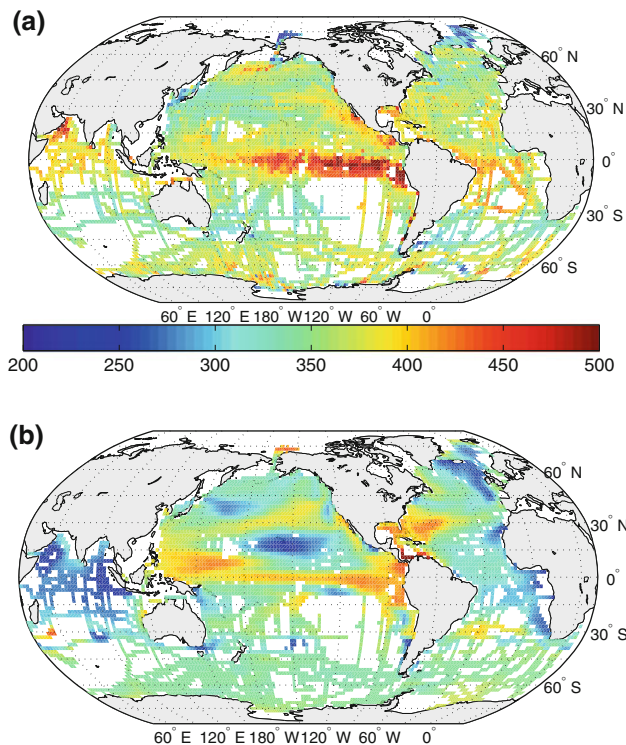


Fig. 6 Maps of 2×2 degrees binned data of surface $p\text{CO}_2$ (μatm) from **a** LDEO dataset (Takahashi et al. 2009), **b** annual climatology of 20C model results

5 Regional changes: projections

5.1 Ocean properties

The projected changes in SST indicate a generalized increase in the northern hemisphere in both scenarios at the end of the twenty-first century (Fig. 4c, d). The SST changes in the A1B are comparable with the changes reported in the IPCC AR4 (Meehl et al. 2007). The highest increases are found in the central north Pacific, North Atlantic and equatorial Pacific, with warming localized in the easternmost and westernmost parts. The changes in the E1 scenario at the end of this century are much lower, especially in the tropical Pacific Ocean, where in most of the basins they are below 2°C . For both A1B and E1, the tropical Pacific warming is somewhat narrow, only limited to the equatorial band. The separation between the eastern and the western parts is still visible in the E1 scenario.

The SST increase in the north Atlantic is instead similar in the E1 and A1B scenarios at the end of the century. This may suggest that in this specific model configuration and spatial resolution, the North Atlantic dynamics are mostly affected by the heating rates that occurred during the twentieth century and the differences in the anthropogenic radiative forcing induced by the GHG pathways do not

further increase the ocean surface temperature in this region (while this occurs at the global scale as shown in Fig. 2).

Figure 7a presents the time evolution of the ocean heat content computed over the uppermost 300 m in the northern Atlantic and Pacific Oceans. Both regions show the initial increase in heat content in the E1 scenario with respect to A1B that was found in the global surface temperature (Fig. 2) and which is attributable to the difference in aerosol forcing between the scenarios at the beginning of the twenty-first century (Fig. 1a). It is shown that the ocean model is particularly sensitive to this factor and that the initial heating of the northern Atlantic in E1 leads to a similar heat content at the end of the twenty-first century as the one found in A1B. Therefore, the similarity in the mean SST state of the northern Atlantic (Fig. 4) is mostly due to the choice of the end-of-century averaging windows, because it is clear that E1 and A1B would diverge as in the northern Pacific Ocean if we would consider a later 30 year period for the comparison.

The CO_2 fluxes with the atmosphere are controlled by the mixing state of the ocean and, especially in the northern Atlantic, by the deep water formation processes. The winter mixing is also a good proxy for the resupply of pre-formed nutrients to the surface ocean and the potential magnitude of the biological carbon pump that will be analysed in the next sections. Figure 7b, c present the projections of change in the March mixed layer depth (MLD) for the northern Atlantic and Pacific and the intensity of the Atlantic MOC at 25 and 48°N for the two scenarios. The MLD is projected to decrease in both scenarios with a larger reduction in E1 at the beginning of this century in combination with the higher heating rate (Fig. 7a). The trends at the end of the century are different for the two basins, with a linear decrease in A1B relative to E1 in the northern Atlantic and an adjustment around values lower than 100 m in the Pacific for both scenarios.

The sensitivity of the model to the prescription of anthropogenic forcings at the beginning of this century is particularly evident in the maximum transport of the Atlantic MOC, shown in Fig. 7c at two different latitudes. The model results indicate a reduction of about 20% at the end of the century in A1B that is in line with the values from other coupled simulations (Schmittner et al. 2005; Crueger et al. 2008). The interesting result is that both scenarios lead to the same MOC values at the end of the century, although this is obtained by a larger initial slow down of the thermohaline circulation that eventually decreases in E1 and by an almost linear decrease over the twenty-first century in A1B.

We note that the maximum MOC transport shows a decadal variability that is also found in the Atlantic MLD.

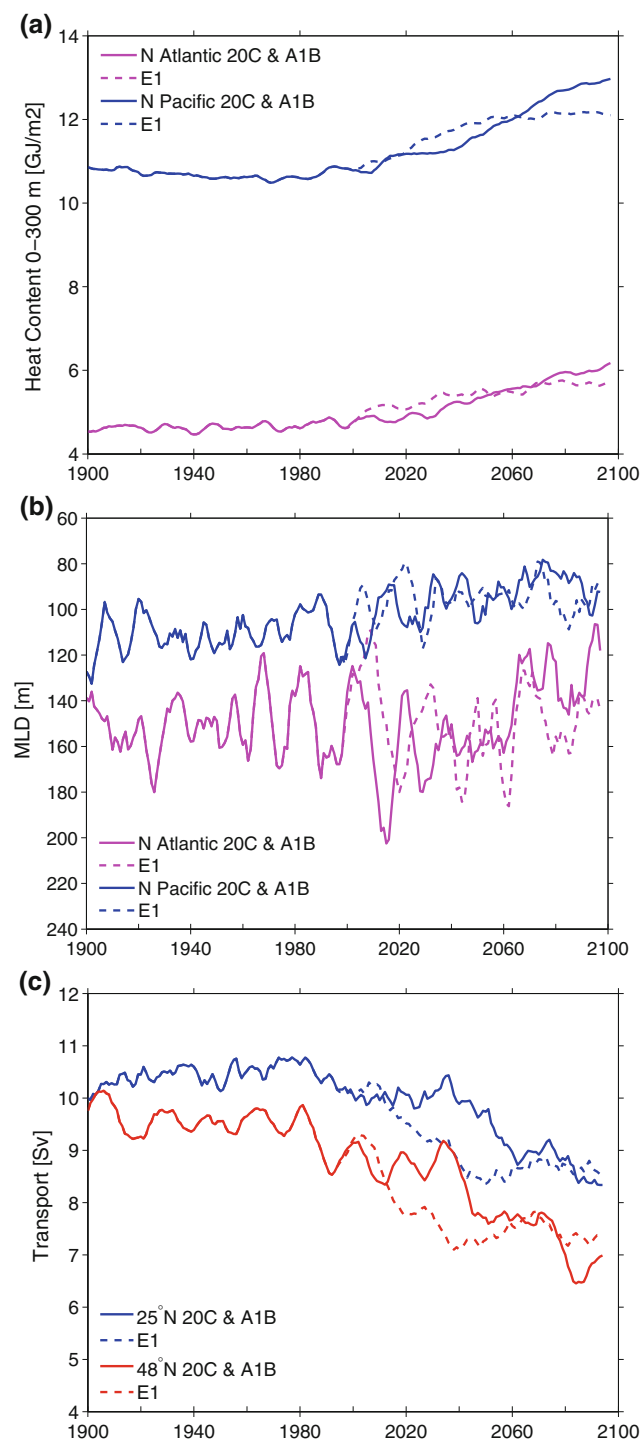


Fig. 7 Annual means of **a** heat content in the uppermost 300 m and **b** March mixed layer depth in the North Atlantic and North Pacific (20°N–60°N) for the 20C and scenario simulations (note that in **b** the vertical scale is reverted). **c** Mean maximum intensity of the Atlantic MOC at 48°N and 25°N for the 20C and scenario simulations. Results are shown as 5-year running means

This timescale of natural variability is observed in other coupled models (e.g. Danabasoglu 2008) and in idealized model studies (Farneti and Vallis 2009).

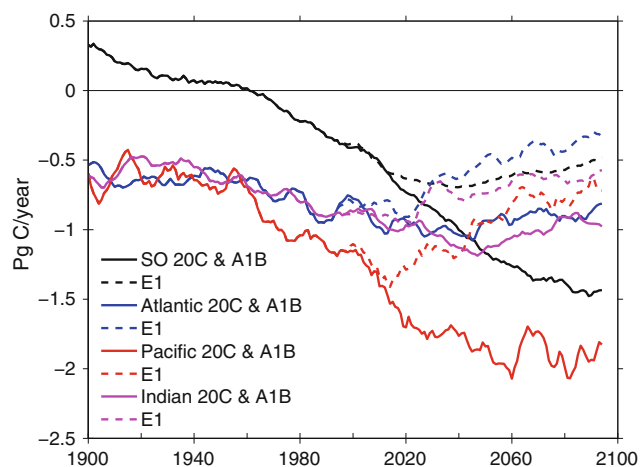


Fig. 8 Ocean carbon uptake in the 20C, A1B and E1 simulations divided into major basin contributions (11 years running means, negative values mean ocean uptake). The Southern Ocean (SO) is defined as the area southern of 50°S. The range of estimates for contemporary net CO₂ fluxes from Gruber et al. (2009) are shown as vertical bars with the relative colour for each basin

5.2 Air–sea carbon fluxes

This section presents the distribution among the major ocean basins (11-years running means in the Southern Ocean south of 50°S, Atlantic, Pacific and Indian oceans, Fig. 8) of the simulated global ocean sink presented in Fig. 3b. The Pacific is the largest ocean sink in the model owing to its large spatial extension, but also because most of its area is characterized by a higher buffering capacity for CO₂ uptake (Feely et al. 2004). In the A1B scenario, the Pacific Ocean responds more rapidly than other basins to the changes in GHG forcings and also shows the highest decadal variability. The ocean carbon uptake increases until 2040–2050 for both the Atlantic and the Pacific Ocean, while the Southern Ocean is still acting as a carbon sink at the end of this century.

The decrease in the ocean uptake is clearly visible in the E1 scenario owing to the strong reduction of atmospheric CO₂. The response is similar in the different basins but the timescales give indications on the basin that is likely to respond faster to the mitigation actions. The Pacific Ocean shows a clear change in the curve derivative prior to 2020 and decrease relatively fast, while the Atlantic is characterised by an earlier onset, longer plateau and a slower uptake reduction.

Under the E1 scenario, the air–sea carbon fluxes in the Pacific at the end of the century are projected to decrease to the levels of 1960 whereas this is not found for the Atlantic and the Southern Ocean. The Atlantic is apparently faster than the Southern Ocean in releasing the anthropogenic carbon to the atmosphere, as the uptake is lower than the value found in the 1960s. The time scales of the Southern

Ocean are longer than the other basins, as it acts as a long-term sink in the A1B scenario and has the slowest tendency of uptake reduction in the E1 scenario.

Figure 8 also shows the range of estimates for contemporary net CO₂ fluxes from Gruber et al. (2009). The overestimation of the contemporary total ocean carbon uptake (Fig. 3) is attributed to biases in the Indian Ocean and partly in the Pacific Ocean, which are related to the biases in surface pCO₂ discussed in Sect. 4.3.

The average patterns of air–sea CO₂ annual fluxes for the last three decades of the 20C and differences with the scenario simulations are shown in Fig. 9. The pattern is mainly driven by changes in the surface values of the ocean pCO₂ (cf. the twentieth century values in Fig. 6b). The reduction of ocean uptake found in the A1B scenario in the Pacific ocean (Fig. 8a) is mostly due to changes in the northern hemisphere where the outgassing is enhanced (Fig. 9b). The largest difference is observed in the northern subtropical gyre below 30° which leads to a change of sign with respect to the 20C simulation and a subsequent extension of the positive region. Ocean uptake increases in the north-western Atlantic and Pacific and in most of the Southern Ocean.

The most striking difference between the scenarios and the 20C simulation is the increase of ocean uptake in the equatorial Pacific both in A1B and E1. This difference is larger than the background value in 20C and therefore it leads to a reversal of the flux in A1B and to values about zero in E1. This change is intuitively in accordance with the increase of SST in the equatorial region (Fig. 4), that is associated to an increase of stratification and a reduced upwelling of colder and DIC-rich waters in the eastern equatorial Pacific (mean vertical velocity in the surface 100 m is reduced of 20% in the A1B and 10% in the E1 scenario, see also Fig. 12). However, the surface DIC value is expected to equilibrate with the atmosphere in the climatological mean, therefore the presence of a mean ocean uptake (i.e. a negative flux) is likely to be related to an enhancement of the physical or biological pump as it will be further discussed in the next section.

The areas of coastal upwelling along the Antarctic continent are characterized by a further decrease of the negative air–sea CO₂ flux, which is much larger in A1B than in E1. Another large increase of uptake is observed in the sub-Antarctic region in correspondence with a southward shift and intensification of the westerlies (a feature found in other coupled climate model results and also in the last 40 years of observations, e.g. Toggweiler and Russell 2008). The E1 scenario (Fig. 9c) presents the same pattern of changes but with a generalized reduction in intensity, apart from the northern North Atlantic and the southern hemisphere mid-latitudes that are more similar in both scenarios.

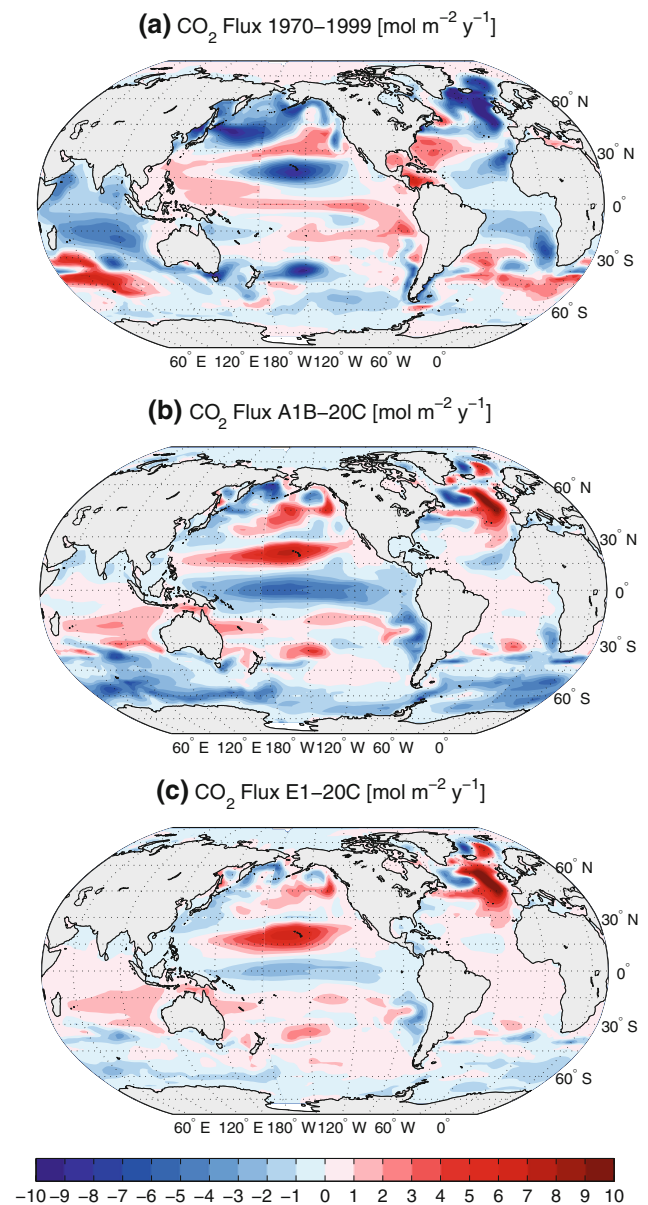


Fig. 9 Annual climatologies (30 years averages) of the surface carbon fluxes to the atmosphere (positive upward). **a** 20C simulation (1970–1999); **b** differences with A1B (2070–2099) and **c** differences with E1 (2070–2099). A negative difference indicate an increase in uptake with respect to the reference simulation

5.3 Changes in the biological carbon pump

The changes in the surface ocean carbon dynamics have been investigated by comparing air–sea CO₂ fluxes and Net Community Production rates (NCP) in three sub-regions of the Pacific and Atlantic oceans (Figs. 10, 11, respectively).

The ocean reverts the net positive CO₂ flux in the tropical Pacific in the last 50 years of the A1B simulation (Fig. 10a), while it is always negative (i.e. ocean uptake) in the northern Pacific and in the southern subtropical gyre. The model indicates that in the last part of the twentieth century

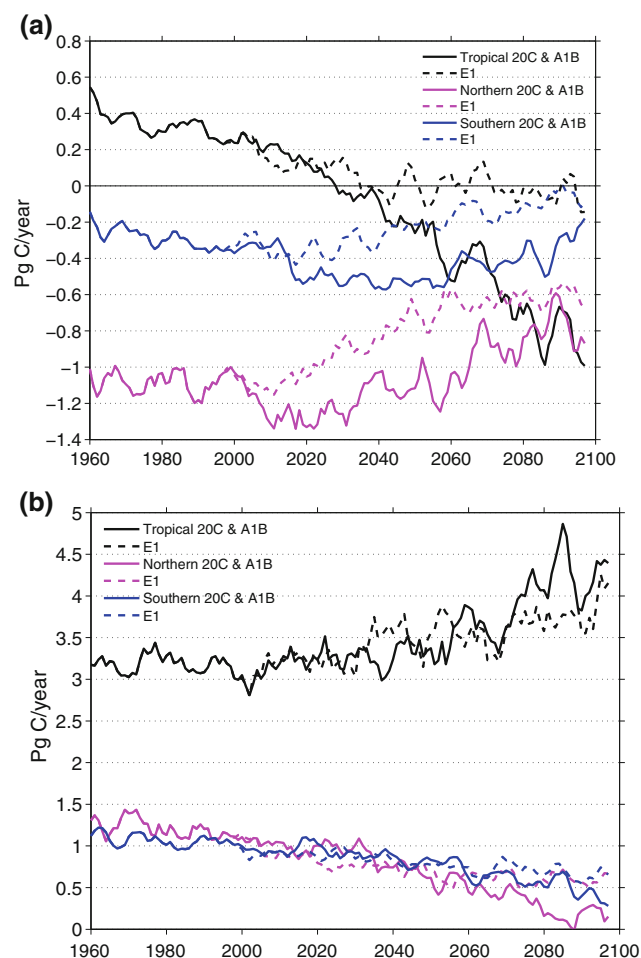


Fig. 10 **a** Net air-sea carbon fluxes in the Pacific Ocean for 20C and scenarios and **b** Net Community Production (NCP) in the euphotic layer for the same simulations. Regions are defined as: Tropical (20°N–20°S); Northern (20–60°N), Southern (20–45°S). Results are shown with 5 years running means, and negative values in **a** correspond to ocean uptake

the outgassing related to the upwelling of cold, DIC-rich waters should have already decreased as a consequence of the increase in SST and reduced upwelling in the eastern equatorial Pacific (Fig. 4), whereas the other sub-basins in the Pacific oceans show lesser changes. From 2030 till the end of the century, the flux in the equatorial Pacific is projected to become a net sink for atmospheric carbon in the A1B scenario (Fig. 10a) and oscillates about zero in the E1 case. Both in the northern and southern regions the A1B and E1 scenarios show an increasing trend towards positive values relative to 20C (i.e. decreased ocean sink). This feature is driven by two different processes as it will be explained further on in the case of the Atlantic Ocean.

In the tropical Pacific, air-sea CO₂ fluxes were found to be inversely correlated with Net Community Production in the surface euphotic zone (Fig. 10b) with a linear correlation coefficient of -0.76 in A1B and -0.66 in E1. NCP is

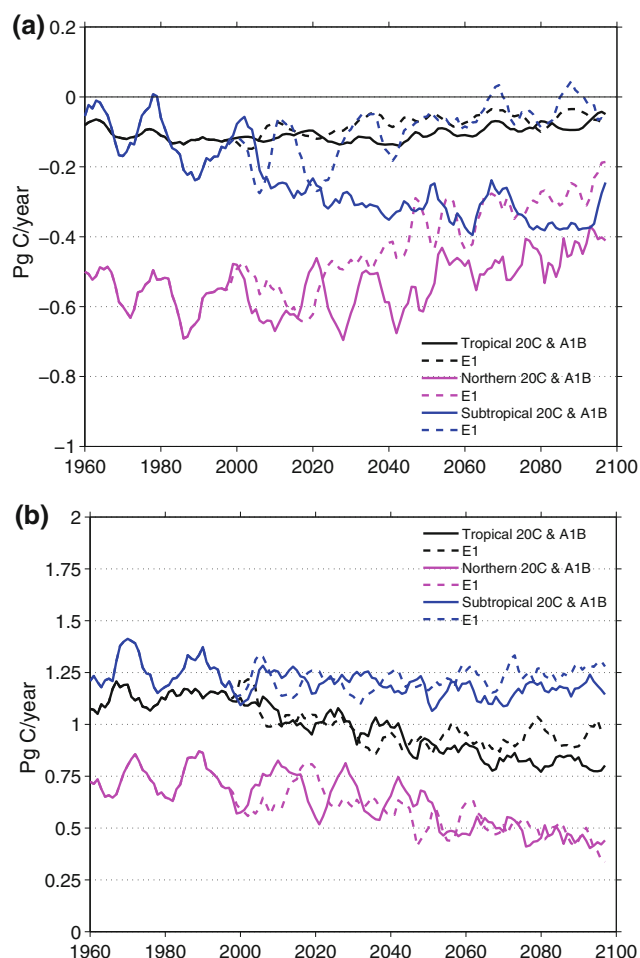


Fig. 11 **a** Net air-sea carbon fluxes in the Atlantic Ocean for 20C and scenarios and **b** Net Community Production (NCP) in the euphotic layer for the same simulations. Regions are defined as: Tropical (20°N–20°S); Northern (45–75°N), Subtropical (20–45°N(S); northern and southern hemispheres are summed together as the trend is the same). Results are shown with 5 years running means, and negative values in **a** correspond to ocean uptake

defined as the difference between primary production integrated over the euphotic zone and the respiration of the planktonic community (both phytoplankton and heterotrophic organisms such as bacteria and zooplankton). Positive values of NCP decrease the surface concentration of DIC and can be considered a proxy for the biological “soft-tissue” pump efficiency. If we assume that all NCP is converted into exportable particulate organic carbon, we have a maximum figure of the potential flux from the surface to the deeper layers where organic consumption processes are slower. The change in CO₂ ingassing is roughly equivalent to the NCP change, indicating that the NCP change in the tropical Pacific could quantitatively account for the air-sea CO₂ flux change in the considered scenarios.

The potential biological pump in the northern Pacific (Fig. 10b) is projected to decrease more than in the

southern Pacific and in the Atlantic (cf. below and Fig. 11b). This is due to a decrease in nutrient supply owing to an increase of vertical stratification (Fig. 7b), a trend that has been documented over the last 30 years in the subarctic north Pacific (Ono et al. 2008). Ono et al. estimated a shrinkage of the productive region of the subarctic Pacific of about 25% by the end of this century considering the linear trend derived from the current observational data. The model estimates that a 25% reduction of NCP with respect to the mean value of 20C would occur as early as year 2020, faster in the case of the E1 scenario than in A1B as it is found for heat content and MLD (Fig. 7a, b). NCP decreases by 50% in the E1 scenario at the end of the century and by 80% in A1B, in this case almost suppressing net community production of lower trophic levels in the euphotic layer. It is interesting to note that changes in net primary production are instead lower (14% in E1 and 20% in A1B) indicating that carbon fixation would only partly be affected whereas the reduced supply of nutrients would undermine the organic particle production which is at the base of the grazing food web. The north Pacific is thus projected to attain a neutral trophic state or even become net heterotrophic at the end of the twenty-first century under both a mitigation and a business-as-usual scenarios.

The Atlantic Ocean, particularly in the northern part of the basin (Fig. 11a), attains about half of the North Pacific carbon sink but with a larger decadal variability. The tropical surface CO₂ flux, differently from the Pacific, is less sensitive to the climate change scenarios. Given the fact that this region shows an unrealistically low surface pCO₂ and a lack of CO₂ outgassing in the contemporary ocean (Figs. 6, 9), it is not possible to assess if this lack of response is due to a realistic process or to the bias in the mean state.

The uptake in the North Atlantic is reduced relative to historical period both in A1B and E1 scenarios but for different reasons. In A1B the decrease in the meridional overturning circulation caused by the tropospheric warming (see Sect. 5.1) diminishes the ocean capability to store the atmospheric carbon, thus reducing the uptake in spite of the increasing atmospheric pCO₂. The air–sea CO₂ flux in E1 (Fig. 11b) is instead controlled by the reduction in atmospheric concentration, and consequently the ocean uptake diminishes.

The reductions in NCP in the northern part of the Atlantic basin (by about 40% in both scenarios relative to the contemporary period) are more similar to the Pacific Ocean and independent of the atmospheric CO₂ concentration pathway. This is linked to a similar change in the water column stratification under the A1B and E1 scenarios evidenced by the same variation in heat content found at the end of the century (Sect. 5.1; Fig. 7a). This increase

leads to a shallower mixed layer depth in the North Atlantic relative to 20C (Fig. 7b) that reduces the mean surface availability of nutrients. The model thus predicts that the E1 mitigation scenario is ineffective in controlling the reduction of NCP, which is expected to be as large as in the Pacific Ocean. This change will however have little impact on the surface carbon fluxes given the fact that in this region the exchange is mostly driven by ventilation processes that are similar in the two scenarios (Fig. 7c).

6 Discussion

The atmospheric CO₂ concentration is the result of the natural and anthropogenic fluxes within the Earth system. The prescription of atmospheric CO₂ in this model of climate change made possible to assess how the ocean adjusts according to internal dynamics and biogeochemistry. Instantaneously, this approach may create local imbalances because of the absence of feedbacks, but in the longer terms (from years to decades) the ocean carbon fluxes are likely to represent the actual oceanic contribution to the pool. This is more likely to be verified when the external forcings are stronger, as it occurred in the second half of the twentieth century with the rise of anthropogenic emissions from fossil-fuel combustion and cement manufacturing (Canadell et al. 2007). When anthropogenic carbon emissions are low, natural variability is dominant and the reconstruction of implied fluxes is less valid.

The results shown here, obtained with a full carbon cycle ESM, indicate that the presence of more refined physical and biogeochemical ocean processes would lead to permissible emissions that are lower than estimated with simpler IAMs, more with the A1B than with the E1 mitigation scenario. This occurs because the oceanic regions respond differently to the forcings, as shown in Sect. 5 by comparing changes in ocean physics with changes in the net air–sea CO₂ fluxes and the biological carbon pump identified by the NCP rate.

One of the most striking results of this modelling work is the change from ocean CO₂ outgassing to ocean uptake in A1B or near equilibrium in E1 in the equatorial Pacific (Fig. 9), which is due to increased NCP (Fig. 10b). These model results suggest that the tropical Pacific is likely to be less vulnerable to future climate change than previously speculated by extrapolating results from physics only coupled climate models and data (e.g. Vecchi et al. 2006).

Generally, scenario simulations (including this work), indicate an equatorial warming in the eastern Pacific and a reduced east–west SST gradient (e.g. Vecchi and Soden 2007). How can the increase in biological net community production observed in the equatorial Pacific be reconciled with a generalized increase of SST (as shown in Fig. 4) that

is intuitively linked to more El Niño-like conditions and thus to a reduction of primary production?

It first needs to be considered that the equatorial circulation is more complex and models do not agree on the ENSO response to the GHGs increase, indicating that the reduction in the gradient does not necessarily lead to the dominance of one or the other of the ENSO states (Vecchi and Soden 2008; Guilyardi et al. 2009). In the INGV-CMCC-CE model there is insignificant change in ENSO variability (not shown) but there is a detectable change in other features of the equatorial ocean circulation associated to the horizontal structure of the SST gradient (Fig. 4). The differences in mean vertical velocity between the scenarios and the contemporary ocean present an increase of the upwelling velocity in the central Pacific and a decrease in the easternmost part (Fig. 12) following a westerly wind anomaly over this latter area (not shown).

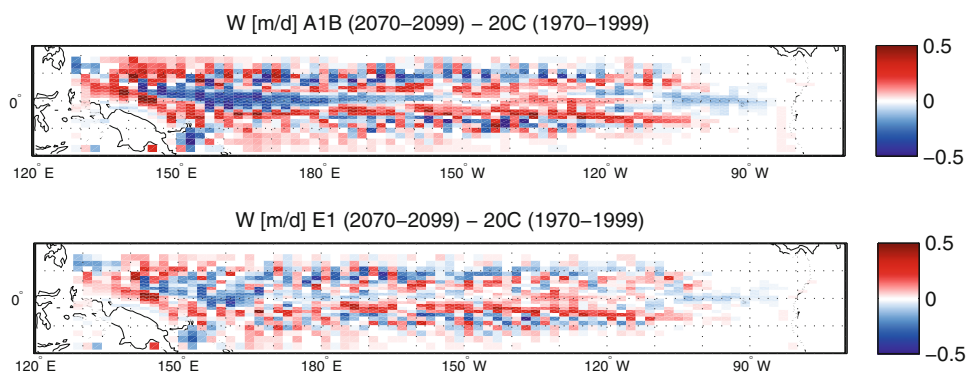
This feature is proposed to be the mechanism responsible for the increase in NCP despite the decrease in upwelling velocity in the other parts of the equatorial Pacific consistently with the surface ocean warming. Upwelling rates in the central equatorial Pacific are connected to subsurface water circulation and to the equatorial undercurrent (EUC, see for instance Weisberg and Qiao 2000). Vichi et al. (2008), by using the same ocean biogeochemistry model of this work, demonstrated that the EUC is the major source of iron to the central and eastern Pacific and that substantial concentrations of iron are found in the core of the EUC in agreement with observations (Coale et al. 1996). Figure 13 presents the timeseries of simulated EUC water mass properties at 140°W, where the EUC waters are defined as the model grid points comprised between 3°S and 3°N and 0–350 m depth that have an eastward velocity component. The modelled transport of the contemporary ocean is in good agreement with the estimates of 25–35 Sv reported by Lukas and Firing (1984) and Sloyan et al. (2003). EUC transport increases by about 4 Sv over the twenty-first century, with little differences between the two scenarios. The weighted transport of DIC (Fig. 13b) evidences the role of ventilation and air–sea CO₂ fluxes in regulating the

amount of carbon flowing in the EUC. The DIC concentration found in the EUC increases linearly in the A1B scenario while it reaches a plateau after 2040 in the E1 case following the prescribed changes in atmospheric pCO₂ (Fig. 1a). The iron transport is instead similar under both scenarios (Fig. 13c, differences between scenarios are not statistically distinguishable), showing a net increase from the values reported in Coale et al. (1996) to concentrations that are 10% higher. Phosphate (Fig. 13d), as well as nitrate (not shown), instead decreases in the EUC, indicating that the EUC waters progressively become enriched in iron and depleted in nutrients. It is well known that the EUC waters originate from subduction in extratropical regions with time-scales ranging from 5 to 100 years (e.g. Gu and Philander 1997; Goodman et al. 2005). This was also demonstrated with the same ocean model and spatial resolution used in this work (Rodgers et al. 2003). Therefore, it is suggested that the increase in EUC transport driven by the climate change scenarios brings more waters into the equator from the subtropical regions, which are characterized by high iron and low macronutrient concentrations like the ones we find in the core of the EUC.

Given the high macro-nutrient concentrations typical of this region, the increase in iron availability through upwelling of EUC waters enhances primary production (Vichi et al. 2008) and a more efficient community production (Sect. 5.3), as it is summarized in Table 1 showing model averages from a region in the central equatorial Pacific from 130° to 150°W. Vertical velocity increases in both scenarios, with an increase in iron inventory and production rates, while macronutrients are reduced although not to limiting levels. The changes in the ratios between nutrients and carbon are not linear as the biogeochemical model parameterizes variable nutrient cell quota in the food web (see Vichi et al. 2007a).

This anomalous increase of the biological pump in the equatorial Pacific has a mechanistic explanation in the framework of the INGV-CMCC-CE model. It will be of interest to see if the current results can be reproduced by other modelling studies, especially comparing with models

Fig. 12 Differences in mean vertical velocity (m/d) in the tropical Pacific (20°S–20°N) between the scenarios (A1B top and E1 bottom) and the 20C simulation. Averaging periods are given in the titles



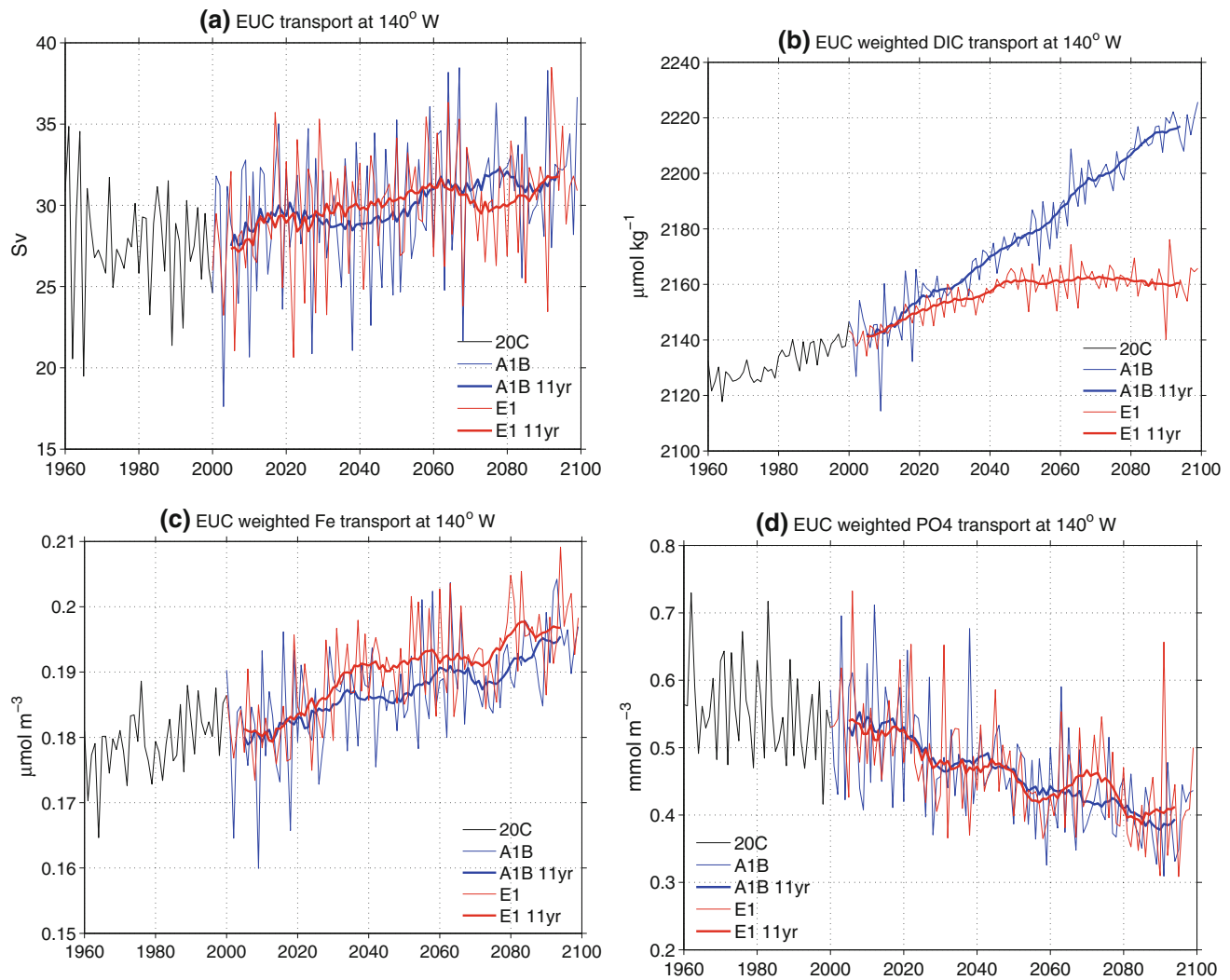


Fig. 13 Changes in the Equatorial Under Current (*EUC*) properties at 140°W: **a** mass transport, **b** mean advected DIC concentration, **c** mean advected dissolved iron concentration, **d** mean advected phosphate concentration

Table 1 Mean values and changes of physical and biogeochemical variables in the central equatorial Pacific (20C 1970–1999; A1B and E1 2070–2099, 120–150°W; 3°S–3°N)

Variable	20C	A1B (% change)	E1 (% change)
w (m/day)	0.84	0.93 (+11%)	0.90 (+7%)
NPP (Pg/year)	6.0	6.4 (+6%)	6.2 (+3%)
NCP (Pg/year)	2.3	3.7 (+61%)	3.3 (+43%)
diss. Fe ($\mu\text{mol m}^{-2}$)	10.9	13.7 (+26%)	13.5 (+24%)
NO ₃ (mmol m^{-2})	129.6	106.2 (−18%)	110.4 (−15%)
PO ₄ (mmol m^{-2})	8.79	7.13 (−19%)	7.60 (−14%)

Net primary production and net community production are computed for a larger region as the one shown in Fig. 10

that uses a more simplified description of biogeochemical processes. Primary production rates in this model have been thoroughly assessed in forced simulations (Vichi and Masina 2009), particularly in the equatorial Pacific, where

it has been shown that model derived rates are equivalent to the estimates from the satellite-based model estimates. The coupled ESM has a generalized lower net primary production at the global scale than found in the forced simulations (NPP in INGV-CMCC-CE is 31.6 Pg/y, NPP in forced run is 46.5 Pg/y), although these results compare well with other ESMs (data from four different models range from 23.7 to 30.7 Pg/y, Schneider et al. 2008). The same models have also been used to estimate the changes in NPP under a climate change scenario (SRES A2, Steinacher et al. 2010). Only one out of four models report an increase in NPP in the equatorial Pacific as found in our results, while at the global scale all models agree that NPP is projected to decrease by about 13% (8% in the E1 and 11% in A1B in our case) but a larger 20% decrease is estimated for export production, which is comparable with our estimates of NCP that is projected to globally decrease by 10% in E1 and 17% in A1B.

The reported regional analysis points out that NCP in the northern Atlantic and Pacific are projected to decrease more than in other regions. The amount of organic carbon made available to the higher trophic levels is projected to be halved in the next 100 years also under the mitigation scenario, with a reduction up to 80% with almost zero net community production from the lower trophic levels of the northern Pacific in the case of the A1B scenario.

In the limit of the formulation of the ocean model used in this work, the fact that these biological responses are of large magnitude also in the case of a substantial mitigation scenario indicates that the simulated changes occurred in the ocean circulation over the twentieth century and the initial part of the twenty-first century determine the response of biogeochemistry in the distant future. The similar response is ascribed to the difference in the aerosol forcing in the two scenarios and to the slow time scales of oceanic processes. The lowest sulphate burden prescribed in E1 relative to A1B before 2050 leads to an initial larger warming of the surface ocean that also involves a larger slow-down of the thermohaline circulation. It is clear from Sect. 5.1 that at the end of the century the time rates of change of surface heat content and mixing state are much larger in A1B than in the E1 scenario, which instead shows a stabilization of the transport in the Atlantic MOC. In the longer term, the E1 scenario is thus projected to achieve the prescribed target, but these results point out that the changes due to cumulative carbon emissions up to present and the projected concentration pathways of aerosol in the very next decades are more relevant for the evolution of surface ocean biogeochemistry than the end of century pathways of atmospheric CO₂ concentrations.

7 Summary and conclusions

The computation of implied fluxes with the INGV-CMCC-CE model over the twentieth century have shown that simulated global carbon fluxes fall within the observed anthropogenic emissions from fossil sources and the estimates of land-use changes. The simulation experiments indicate that the ocean uptake of anthropogenic CO₂ emissions is projected to reduce during the twenty-first century in case of the A1B scenario of socio-economic development. The projections are however different under the strong constraints of an aggressive scenario of emission mitigation like the ones prescribed in the E1 case. In this case, ocean uptake at the end of this century is expected to return to the same magnitude of air–sea carbon fluxes as simulated in 1960.

Model projections indicate that the presence of a dynamical carbon cycle in the coupled climate system requires a reduction of the permissible anthropogenic emissions of about 20% with respect to the ones assumed

in the A1B scenario and in the 450 ppm stabilization scenario E1. The carbon mitigation rates prescribed in the E1 scenario are likely to achieve the targeted CO₂ pathway, also considering the dynamical response of the natural carbon cycle implemented in the model.

In both cases, the ocean capacity to uptake atmospheric carbon is not uniformly distributed over the global ocean. While the Southern Ocean is projected to linearly increase the storage of anthropogenic carbon with A1B, the Pacific and Atlantic Ocean are anticipated to show the first signs of saturation in the next 30 years, though some ocean regions such as the tropical Pacific will show an increase in the air–sea CO₂ uptake. The reason for this contrasting response is ascribed to complementary changes in the ocean vertical stratification and the biological carbon pump intensity. While the majority of northern hemisphere basins is expected to feedback positively to the increase of atmospheric pCO₂ because of the decrease in solubility and deep water ventilation, the Southern Ocean and the tropical Pacific will increase the surface uptake rates. In the tropical Pacific, this feature is attributed to the enhancement of the biological carbon pump evidenced by an increase in Net Community Production following changes in the subsurface equatorial circulation and enhanced iron availability in the central equatorial Pacific from sub-tropical regions through the Equatorial Undercurrent.

However, Net Community Production of lower trophic levels in the northern Pacific and Atlantic oceans is projected to decrease substantially (from 50 to 80%) at the end of the twenty-first century. This occurs also in the case of the aggressive 450 ppm mitigation scenario that imposes marked emission reduction starting from 2015 to 2020 but also a marked reduction of sulphate burden that increase the warming of the surface ocean in the first part of this century and affects the subsequent evolution of surface biogeochemistry. Although the NCP decrease appears to have little impact on the ocean carbon uptake capacity at the end of the century, the consequences for marine ecosystems may be potentially large and needs to be assessed with complementary studies that include higher trophic levels beyond the planktonic components used in this work.

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